
Microeconomics I

Class 1 — Preferences

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Who I am

- Studied in Pilsen, Kaohsiung, Barcelona and New York
- Mathematics first, then economics — PhD, New York University, 2019
- Across the Atlantic by sailboat, twice
- Now: the economics of AI safety and governance

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Plan for today

10:00	Why any of this matters
10:20	Diagnostic quiz — not graded
10:30	Part I: the nine sessions, the grade
10:40	The consumption set
11:00	Break
11:10	Answers to the exercise; quiz results
11:20	Binary relations and their properties
11:40	Rational preferences: $\succsim$, $\succ$, $\sim$

The Economist's subscription offer

A

Web only

\$59

B

Print only

\$125

C

Print + web

\$125

Hands up: A, B or C?

The offer as used in Ariely (2008), ch. 1.

An option nobody chose changed the choice

Group 1 $n = 100$	Web 16	Print 0	Print + web 84
Group 2 $n = 100$	Web 68		Print + web 32

Another 100 students, same prices, middle option removed. Did the 84 make a **mistake**?

Ariely (2008), ch. 1; Huber, Payne & Puto (1982).

The cost of living, measured wrong by 1.1 points a year

1.1 pp

per year: how much the US CPI overstated the rise in the cost of living

0.4 pp of it: consumers substitute when prices move

Compounded, it moves pensions and tax brackets. The fix: session 6.

Boskin Commission, report to the US Senate Finance Committee (1996).

What consumer theory lets you do

- Predict how demand reacts to prices and taxes
- Put a euro value on a policy: who gains, who loses
- Test rationality on purchase data, no utility needed
- Price risk: insurance, portfolios, lotteries

BLP (1995) · Hausman (1981) · Samuelson (1938), Afriat (1967) · von Neumann & Morgenstern (1944).

Diagnostic quiz

- Eight questions — **not graded**
- Shows me what to explain and what to skip
- Your name on the sheet, circle the letters
- About ten minutes, then tear off the strip and hand it in

Not seen it before? Leave it blank, do not guess.

Part I in nine sessions

			MWG
1	Tue 29 Sep	Consumption set, binary relations, preferences	1.B, 2.C
2	Wed 30 Sep	Monotonicity, convexity, indifference curves	3.B
3	Tue 6 Oct	Utility representation	3.C
4	Wed 7 Oct	Utility maximisation, Walrasian demand	2.D, 3.D
5	Tue 13 Oct	Indirect utility, price and wealth effects	2.E, 3.D, 3.G
6	Wed 14 Oct	Expenditure minimisation, Hicksian demand	3.E–3.G
7	Tue 20 Oct	Slutsky equation; welfare measures	3.G, 3.I
8	Wed 21 Oct	Uncertainty I: expected utility, risk aversion	6.B–6.C
9	Tue 27 Oct	Uncertainty II: Arrow–Pratt, dominance; review	6.C–6.D

Wed 28 Oct Part I exam

MWG · Muñoz-Garcia, *Advanced Microeconomic Theory*, ch. 1, 2, 5 (gentler).

How Part I is graded

Part I exam, Wed 28 Oct, in class	40 %
Two problem sets, groups of 2–5	5 %
Short quizzes in class, on paper	5 %

Groups of 2–5: sign-up sheet **tomorrow**. Quizzes weekly, closed book, lowest dropped.

Part II: Olga Chiappinelli, from November. Pass: 40 % in every component, 50 % overall.

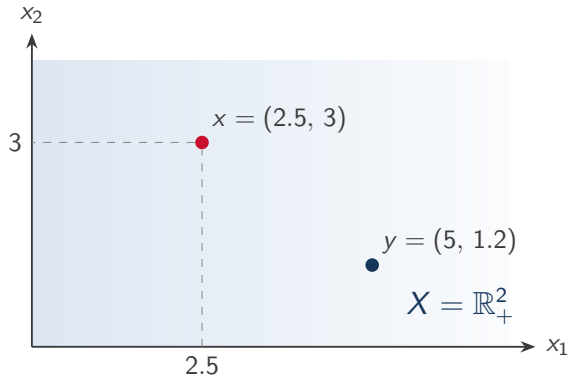
Practicalities

- Course page: pkocourek.com/teaching/micro-1
- Office hours: Wednesdays 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Problem set 1 out Tue 6 Oct, due Wed 14 Oct
- AI: none in quizzes and exams; in problem sets, declare it

The consumption set

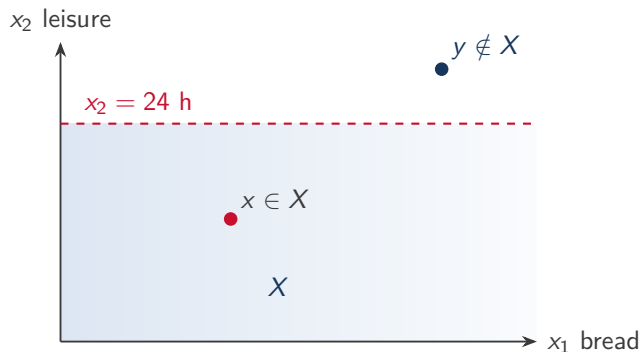
MWG 2.C

Two goods, one point



A bundle is a vector. X is what can be consumed; affordability comes later.

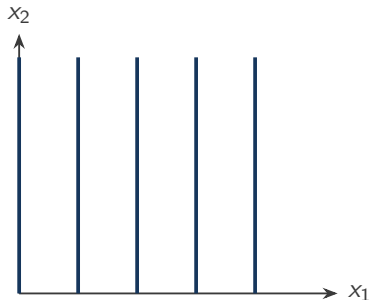
Physical limits cut the set



$X = \{x \in \mathbb{R}_+^2 : x_2 \leq 24\}$. A day has 24 hours.

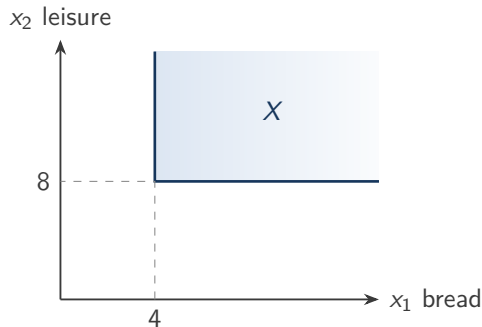
After MWG 2.C, redrawn.

Indivisible units, survival



good 1 only in whole units

$$X = \{x \in \mathbb{R}_+^2 : x_1 \in \{0, 1, 2, \dots\}\}$$



you must eat and sleep

$$X = \{x \in \mathbb{R}_+^2 : x_1 \geq 4, x_2 \geq 8\}$$

After MWG 2.C, redrawn.

In pairs: sketch X

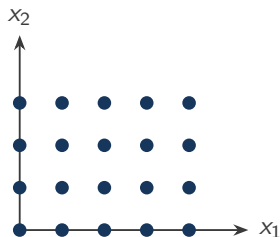
1. Both goods sold only in whole units
2. Bread in Barcelona (x_1) or bread in Madrid (x_2) — you can be in one place
3. Leisure between 8 and 24 hours a day, bread any amount

Three minutes. Which of the three sets are **convex**?

Break

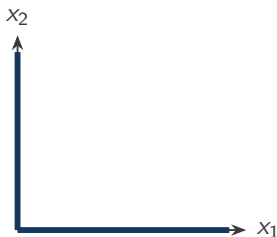
Back at 11:10

Answers: sketch X



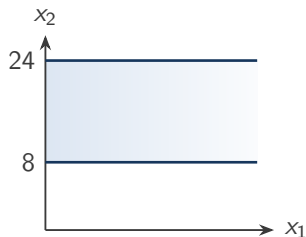
1. $x_1, x_2 \in \{0, 1, 2, \dots\}$

not convex



2. $x_1 = 0$ or $x_2 = 0$

not convex

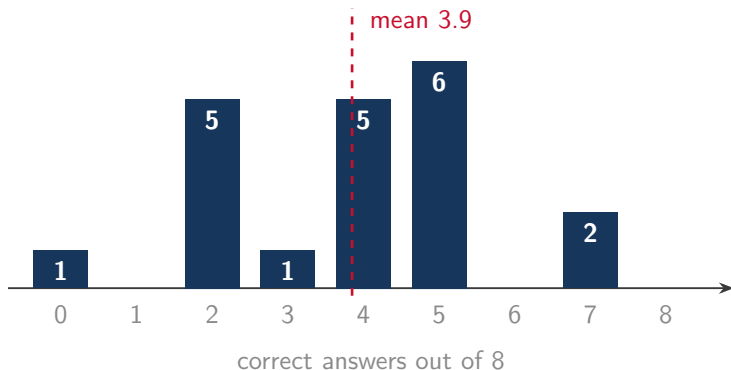


3. $8 \leq x_2 \leq 24$

convex

Convex: the segment between any two points of X stays in X . Only the third set passes.

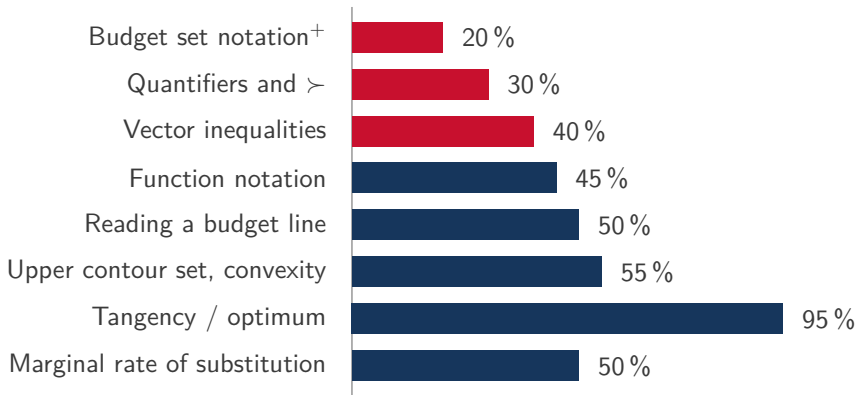
Diagnostic quiz: scores



20 strips. Median 4. Not graded.

A question with several correct options counts only if the whole set is right.

Diagnostic quiz: share right, by question



n = 20. ⁺ select all that apply: the whole set must be right.

Binary relations

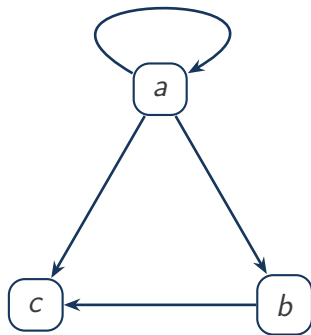
and their properties

Definition: binary relation

A **binary relation** R on X is a set $R \subset X \times X$.

Write $x R y$ for $(x, y) \in R$.

$\geq$ on $\mathbb{R}$ · “is taller than” · $\subset$ on sets



$$R = \{(a, a), (a, b), (b, c), (a, c)\}$$

Properties of a relation, I

complete: for all x, y : $x R y$ or $y R x$ $\geq$ on $\mathbb{R}$

reflexive: for all x : $x R x$ “has the same birthday as”

transitive: $x R y$ and $y R z \Rightarrow x R z$ “is an ancestor of”

Complete $\Rightarrow$ reflexive. Take $y = x$.

Properties of a relation, II

symmetric: $x R y \Rightarrow y R x$

“is a sibling of”

asymmetric: $x R y \Rightarrow \text{not } y R x$

“is taller than”

antisymmetric: $x R y \text{ and } y R x \Rightarrow x = y$

$\geq$ on $\mathbb{R}$

Asymmetric $\Rightarrow$ antisymmetric, and **irreflexive:** never $x R x$.

In pairs: which properties hold?

	complete	reflexive	symmetric	transitive
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$\geq$ on $\mathbb{R}$

$>$ on $\mathbb{R}$

“is a sibling of”

$x \geq y$ in both coordinates, on $\mathbb{R}_+^2$

$x_1 + x_2 \geq y_1 + y_2$, on $\mathbb{R}_+^2$

Four minutes.

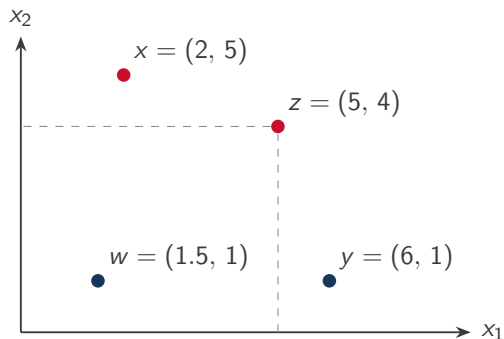
In pairs: which properties hold?

	complete	reflexive	symmetric	transitive
$\geq$ on $\mathbb{R}$	✓	✓	✗	✓
$>$ on $\mathbb{R}$	✗	✗	✗	✓
“is a sibling of”	✗	✗	✓	✗
$x \geq y$ in both coordinates, on $\mathbb{R}_+^2$	✗	✓	✗	✓
$x_1 + x_2 \geq y_1 + y_2$, on $\mathbb{R}_+^2$	✓	✓	✗	✓

Siblings: transitivity is a claim for *all* a, b, c , including $c = a$.

Then $a R b$ and $b R a$ would force $a R a$ — and no one is their own sibling.

Coordinatewise $\geq$ is not complete



$z \geq w$, but x and y are not ranked. A preference must rank them anyway.

Rational preferences

MWG 1.B

The preference relation

$\succsim$ is a binary relation on X : $x \succsim y$ “ x is at least as good as y ”

$x \succ y \iff x \succsim y$ and not $y \succsim x$ strict preference

$x \sim y \iff x \succsim y$ and $y \succsim x$ indifference

MWG 1.B.

Definition: rational preferences

$\succsim$ is **rational** if it is

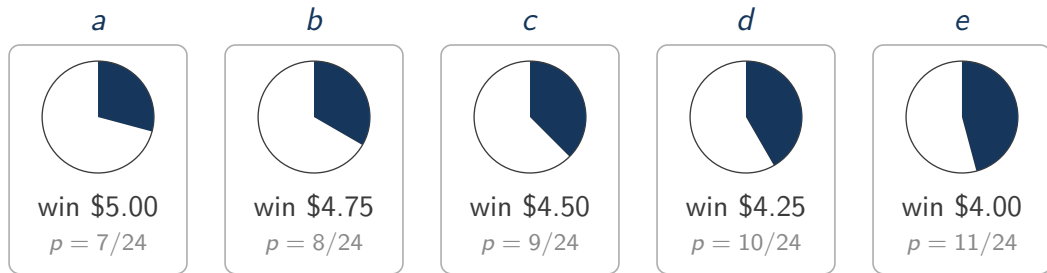
complete: for all $x, y \in X$: $x \succsim y$ or $y \succsim x$

transitive: for all $x, y, z \in X$: $x \succsim y$ and $y \succsim z \Rightarrow x \succsim z$

Completeness: you can always rank two bundles — even ones you have never faced.

MWG Definition 1.B.1.

Five gambles: which do you take?



***a* or *b*? *b* or *c*? ... *a* or *e*?**

Shaded sector: the chance of winning. Subjects saw the discs, not the fractions.

Tversky (1969), *Psychological Review* 76, Experiment I, Table 1.

Transitivity: what people did

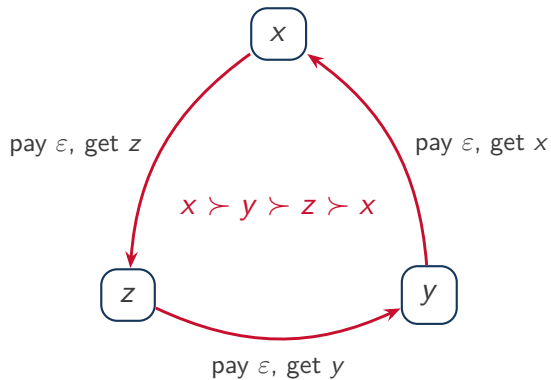
$$a \succ b, \quad b \succ c, \quad c \succ d, \quad d \succ e$$

$$\text{and yet} \quad e \succ a$$

8 of 18 students screened showed this pattern: small probability gaps ignored, big ones not.

Tversky (1969), *Psychological Review* 76(1), 31–48.

Why cycles cost you: the money pump



Same object, 3ε poorer, ready to go round again.

Davidson, McKinsey & Suppes (1955), *Philosophy of Science* 22.

What rationality gives you for free

If $\succsim$ is rational, then

(i) $\succ$ is irreflexive and transitive

hence asymmetric

(ii) $\sim$ is reflexive, symmetric and transitive

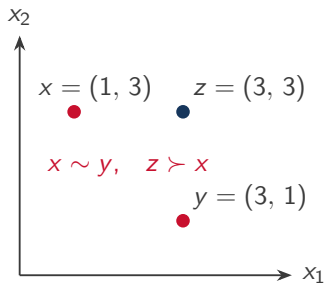
an equivalence relation

(iii) $x \succ y \succsim z \Rightarrow x \succ z$

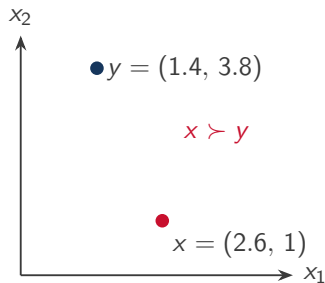
Prove (iii) in two lines. Which of the two axioms does it use?

MWG Proposition 1.B.1.

Two preferences on $\mathbb{R}_+^2$ — rational?



$$x \succsim y \iff x_1 + x_2 \geq y_1 + y_2$$



$$x \succsim y \iff x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

Two minutes with your neighbour: complete? transitive? What do $\sim$ -classes look like?

Back to the decoy

{ Web, Print + web }

most chose **Web**

{ Web, Print, Print + web }

most chose **Print + web**

If each student picks the $\succsim$ -best item, Web vs Print + web is settled before any menu is shown.

Adding Print, which nobody chose, cannot flip it — yet Web went from 68 to 16.

Groups 1 and 2 of the reveal slide. Ariely (2008), ch. 1.

Today in one line



Tomorrow: what indifference curves look like when $\succsim$ is monotone and convex.

Read MWG 3.B (or Muñoz-Garcia ch. 1) for Wednesday. Bring your group of 2–5.